

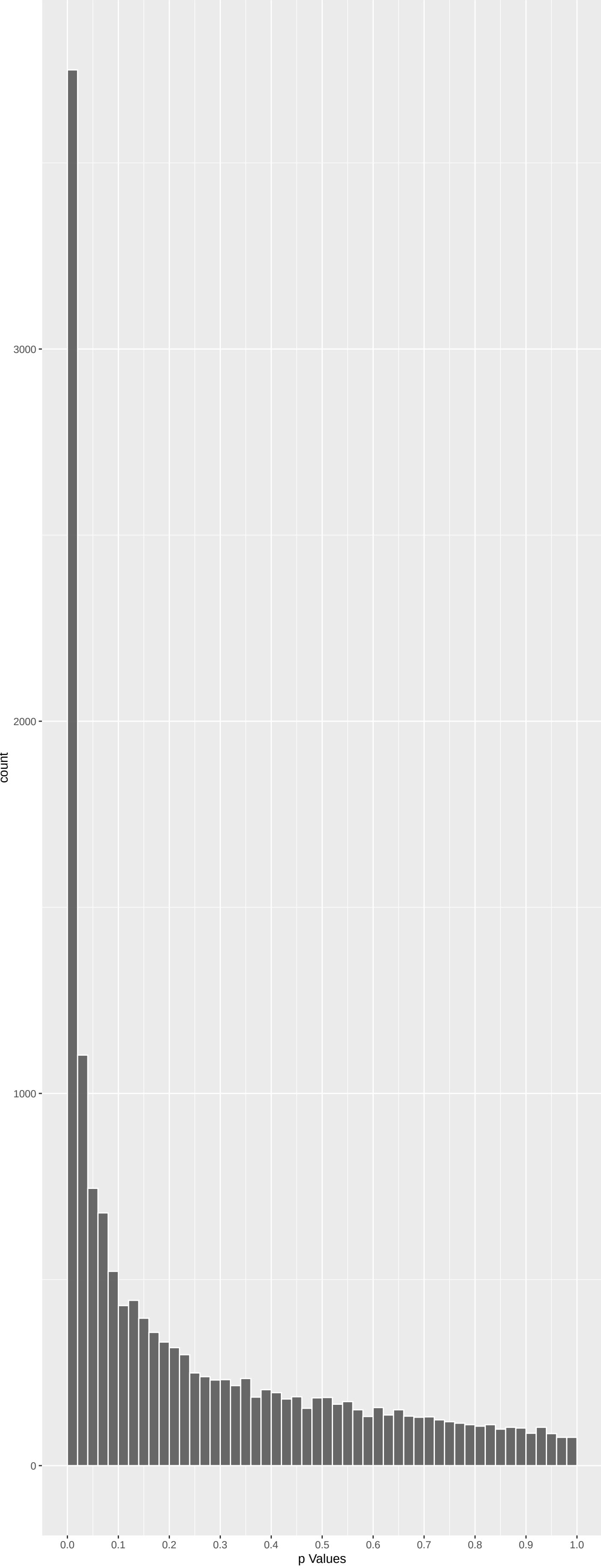
Top genes by P-value non-permulated

Gene	Rho	N	P	p.adj	qValueNoperm
CLDN16	0.2332371	286	2.428e-14	3.590e-10	1.023e-06
CPB1	0.1849700	353	4.753e-14	3.590e-10	1.023e-06
BPIFB1	0.1534732	347	1.744e-11	7.253e-08	1.033e-04
SLC5A12	0.1556035	341	1.921e-11	7.253e-08	1.033e-04
TRPM6	0.1405656	366	4.209e-11	1.236e-07	1.174e-04
ADGRF1	0.1393247	367	4.909e-11	1.236e-07	1.174e-04
MUC20	0.1734823	276	2.457e-10	5.303e-07	4.318e-04
BNC1	0.1393611	332	5.326e-10	1.006e-06	7.164e-04
PGC	0.1557136	294	6.817e-10	1.144e-06	7.246e-04
IQGAP3	0.1241827	360	1.135e-09	1.623e-06	8.410e-04
CPLX3	0.2181356	205	1.182e-09	1.623e-06	8.410e-04
EP400	0.1220749	356	2.088e-09	2.628e-06	1.145e-03
HIRA	0.1540219	280	2.495e-09	2.899e-06	1.145e-03
CLCNKA	0.1322568	324	2.823e-09	3.013e-06	1.145e-03
CFAP20DC	0.1241996	344	2.992e-09	3.013e-06	1.145e-03
CRYAA	0.2354491	180	3.737e-09	3.528e-06	1.211e-03
SLC14A2	0.1165783	361	4.075e-09	3.621e-06	1.211e-03
PROM1	0.1131160	370	4.556e-09	3.823e-06	1.211e-03
CTRC	0.1222035	335	7.154e-09	5.688e-06	1.705e-03
TMEM131L	0.1084702	364	1.211e-08	9.146e-06	2.075e-03
CAMK2B	0.1405314	282	1.367e-08	9.833e-06	2.075e-03
NARS1	0.1248748	314	1.665e-08	1.050e-05	2.075e-03
SPG11	0.1030050	377	1.980e-08	1.050e-05	2.075e-03
SPINT4	0.1402108	278	1.878e-08	1.050e-05	2.075e-03
TMED6	0.1248729	313	1.770e-08	1.050e-05	2.075e-03

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BNC1	0.1393611	332	5.326e-10	1.006e-06	7.164e-04
PGC	0.1557136	294	6.817e-10	1.144e-06	7.246e-04
IQGAP3	0.1241827	360	1.135e-09	1.623e-06	8.410e-04
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SLC14A2	0.1165783	361	4.075e-09	3.621e-06	1.211e-03
CTRC	0.1222035	335	7.154e-09	5.688e-06	1.705e-03
NARS1	0.1248748	314	1.665e-08	1.050e-05	2.075e-03
SPG11	0.1030050	377	1.980e-08	1.050e-05	2.075e-03
SPINT4	0.1402108	278	1.878e-08	1.050e-05	2.075e-03
TMED6	0.1248729	313	1.770e-08	1.050e-05	2.075e-03
TIGD4	0.1234942	315	1.935e-08	1.050e-05	2.075e-03
DST	0.1087756	360	1.693e-08	1.050e-05	2.075e-03

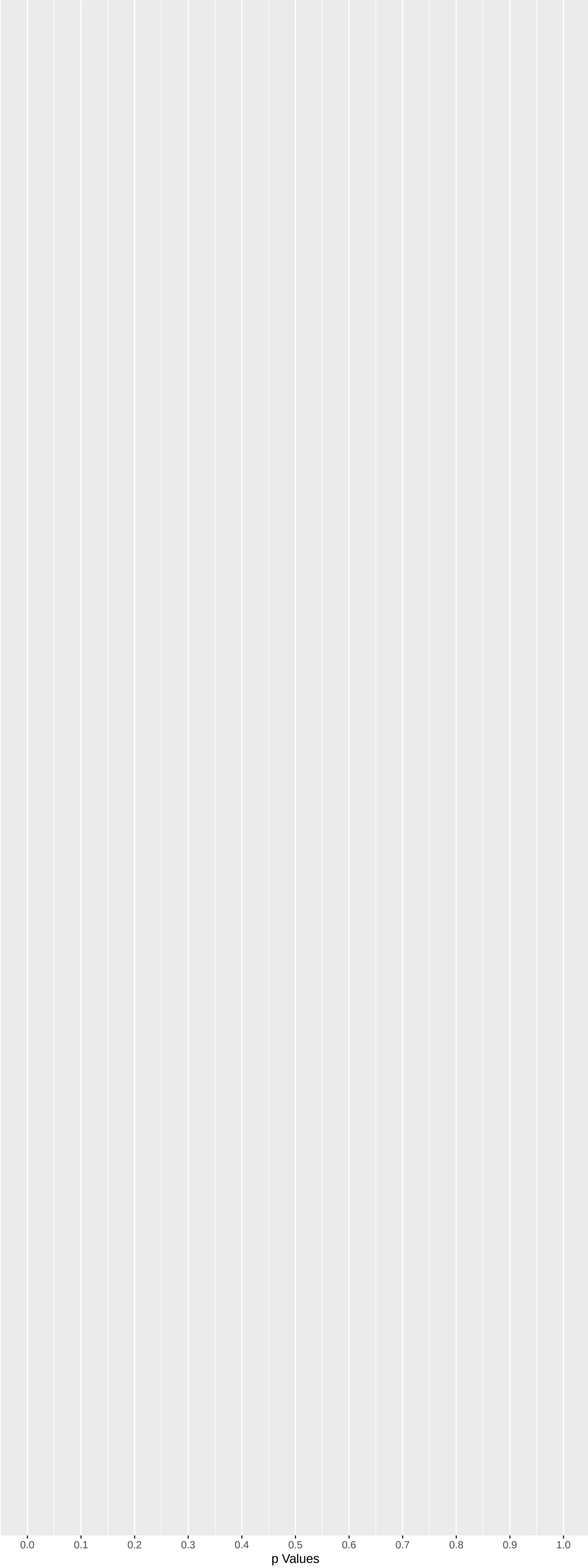
Positive Rho Non-permulated



Top Positive genes by P-value non-permulated

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CPB1	0.1849700	353	4.753e-14	3.590e-10	1.023e-06
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BNC1	0.1393611	332	5.326e-10	1.006e-06	7.164e-04
PGC	0.1557136	294	6.817e-10	1.144e-06	7.246e-04
IQGAP3	0.1241827	360	1.135e-09	1.623e-06	8.410e-04

Negative Rho Non-permulated



Top Negative genes by P-value non-permulated

Gene	Rho	N	P	p.adj	qValueNoperm
NA	NA	NA	NA	NA	NA
NA.1	NA	NA	NA	NA	NA
NA.2	NA	NA	NA	NA	NA
NA.3	NA	NA	NA	NA	NA
NA.4	NA	NA	NA	NA	NA
NA.5	NA	NA	NA	NA	NA
NA.6	NA	NA	NA	NA	NA
NA.7	NA	NA	NA	NA	NA
NA.8	NA	NA	NA	NA	NA
NA.9	NA	NA	NA	NA	NA

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Monocarboxylic Acid Transport (GO:001571	0.25316227	50	6.202e-10	3.331e-06	SLC5A12:15 SLC16A5:222 SLC10A3:282 SLC5A8:439 SLC10A2:474 SLC16A9:568 SLC6A12:596
Organic Hydroxy Compound Transport (GO:0	0.21918957	36	5.435e-06	1.460e-02	SLC10A3:282 SLC5A8:439 SLC10A2:474 SLC22A1:690 ABCG5:834 SLC01B1:860 SLC51B:931
Lysosome Organic Anion Transport (GO:032418	-0.23448608	27	2.500e-05	4.476e-02	SPAG9:18 GRP65:599 NDC14:743 KIT:807 HP56:875 BORC1:1032 SNAPIN:1457
Bile Acid and Bile Salt Transport (GO:000	0.28682788	16	7.157e-05	9.611e-02	SLC10A3:282 SLC10A2:474 SLC01B1:860 SLC51B:931 SLC51A:939 SLC10A6:1062 ABCD3:1405
Organic Anion Cation Transport (GO:0015695)	0.20286781	30	1.420e-04	1.255e-01	SLC22A16:24 SLC4A45:401 SLC6A12:596 SLC22A1:690 SLC18B1:994 RALBP1:1037 SLC19A2:1648
Positive Regulation Of Cell Population P	-0.05836496	328	3.177e-04	2.844e-01	MAPK15:35 GDNF:36 KMT2D:75 IL5RA:84 ZPFM2:99 IGF1R:101 CDH13:121
lRNA Methylation (GO:0030488)	0.17542527	34	4.045e-04	3.103e-01	TARBP1:259 TRMT10A:592 TRMT9B:1242 LCM2:7292 METTL2:189 TRMT10C:1800 TYW3:1813
Carboxylic Acid Transport (GO:0046942)	0.14291867	47	7.095e-04	4.234e-01	SLC5A12:15 SLC16A5:222 SLC36A2:227 SLC6A6:519 SLC36A1:564 SLC16A9:568 SLC6A12:596
Small Nucleolar Ribonucleoprotein Comple	-0.32775028	9	6.632e-04	4.234e-01	TNIP1:139 RUVBL2:704 SHQ1:1577 RUVBL1:1640 Pih1D1:144 ACA2:239 N2H13:259 ZNH13:3045
RNA Methylation (GO:0001510)	0.14016765	48	7.917e-04	4.252e-01	TARBP1:259 TRMT10A:592 FTSJ3:618 TRMT9B:1242 LCM2:7292 CMTR1:1326 RBM15:1359
Energy Derivation By Oxidation Of Organi	-0.20015078	22	1.161e-03	4.372e-01	COX15:42 COX6A1:345 PSARGC1A:1141 COX4I1:1162 CYC1:1463 NDUFS4:1526 NR4A3:1537
Mast Cell Activation Involved In Immune	-0.37570679	6	1.438e-03	4.372e-01	GRP:659 KIT:807 NR4A3:1537 PIK3CD:1596 FCER1A:2348 PIK3CG:2551 NA
Mast Cell Degranulation (GO:0043303)	-0.37570679	6	1.438e-03	4.372e-01	GRP:659 KIT:807 NR4A3:1537 PIK3CD:1596 FCER1A:2348 PIK3CG:2551 NA
Mast Cell Mediated Immunity (GO:0002448)	-0.37570679	6	1.438e-03	4.372e-01	GRP:659 KIT:807 NR4A3:1537 PIK3CD:1596 FCER1A:2348 PIK3CG:2551 NA
Monoatomic Cation Homeostasis (GO:000508	0.11946207	61	1.273e-03	4.372e-01	CYBRD1:66 KCNH2:133 SLC12A8:160 CLDN16:186 LETM1:383 CNNM4:537 TRPV5:578
Neuron Differentiation (GO:0030182)	-0.08695728	114	1.387e-03	4.372e-01	RTN4:132 USH2A:265 SPINK5:358 TENM1:431 ERBB4:453 WNT7B:470 WNT9B:520
Peptidyl--Tyrosine Phosphorylation (GO:00	-0.12634054	57	9.858e-04	4.372e-01	IGF1R:101 ERBB4:453 AB13:473 EFEMP1:593 DDR1:620 RSLN:634 FGR:740
Ubiquitin--Dependent Protein Catalytic Pr	-0.06180434	226	1.465e-03	4.372e-01	AMBR1:34 TBL1XR1:40 PSMA7:138 USP33:180 PSDM7:184 PSF81:191 RNF122:281
Positive Regulation Of T Cell Activation	-0.10627397	74	1.606e-03	4.539e-01	CD276:373 THY1:500 TNFSF11:669 CD274:721 ZP4:784 IL21:845 HLA-DRA:849
Stem Cell Differentiation (GO:0016072)	-0.17755037	26	2.176e-03	4.609e-01	LMBR1:55 TP53:297 FOXO4:499 KIT:807 FGF2:925 MEOK1:1014 SETD6:1090
Calcium--Independent Cell--Cell Adhesion V	0.26764411	11	2.118e-03	4.783e-01	CLDN16:186 CLDN8:211 CLDN16:186 CLDN20:1460 C12orf11:1462 CLDN23:2164 CLDN15:2757
Intracellular Monoatomic Ion Homeostasis	0.20591185	19	1.967e-03	4.783e-01	CLDN16:186 LETM1:383 CNNM4:537 SLC4A1:670 RHAG:914 RHCG:1513 SLC39A10:2648
Lipid Transport (GO:0008695)	0.10030826	78	2.293e-03	4.783e-01	EHDH4D:116 ABCB1:144 ABCA2:161 SLC1A2:160 SLC10A2:474 MTP:506 CHKA:611
Positive Regulation Of Double--Strand Bre	0.28281818	10	2.195e-03	4.783e-01	KMT5C:924 WRAP53:967 SMCHD1:1003 KMT5B:1399 PRKDC:1509 SHLD2:1655 PNKP:3374
Regulation Of Synaptic Vesicle Exocytosi	-0.25404122	12	2.358e-03	4.783e-01	RAP1A:1 VPS18:171 SLMC1:137 SLMC2:136 SLMC3:136 SLMC4:136 SLMC5:136 SLMC6:136
Ribosome Biogenesis (GO:0042254)	-0.08360945	112	2.306e-03	4.783e-01	TNIP1:139 PAK1P1:195 NOL1:372 C1NP:443 WDRT5:463 BSYL4:477 RPSS5:541
Positive Regulation Of Defense Response	0.23902621	7	2.433e-03	4.836e-01	PGC:5 EMILIN2:135 GRN:723 CLK5:1274 CLK7:1275 EMILIN1:5581 MMRN2:6006
Positive Regulation Of Transcription By	-0.03595721	628	2.521e-03	4.836e-01	ESRRG:4 NR5A2:122 GDNF:36 ABHD148:338 TBL1XR1:40 STING1:41 KMT2D:75
rRNA Metabolic Process (GO:0016072)	-0.11060048	62	2.637e-03	4.884e-01	NOL1:372 BSYL4:477 NOP58:748 DDY5A:984 XRN2:1125 ESF1:1134 EMG1:1270
Antigen Receptor--Mediated Signaling Path	-0.09003281	90	3.228e-03	4.917e-01	CD276:373 NFATC2:457 TRAF6:497 THY1:500 STOML2:581 EIF2B2:625 RC3H1:635
Cardiac Right Ventricle Morphogenesis (G	-0.42305169	4	3.387e-03	4.917e-01	ZFPM2:99 GATA4:909 JAG1:900 SOX4:1926 NA NA
Centromere Sister Chromatid Cohesion (G	0.42507066	4	3.238e-03	4.917e-01	MEIKIN:319 SGO1:740 BUB1:1114 BUB1B:1623 NA NA
Ear Morphogenesis (GO:0042471)	0.22680159	14	3.309e-03	4.917e-01	TIFAB:68 USH1C:179 FGF2:955 OSR2:2716 EPHB2:2746 NCG:2781 NEUROG1:2900
Fac-Soluble Vitamin Metabolic Process (G	0.21310895	16	3.174e-03	4.917e-01	CYP11A1:82 CYP2R1:92 CYP11A:318 IRTAT:675 SARGC:1502 KM5A:2026 NQO1:2534
Metabolism Biosynthetic Process (GO:0	0.08955076	104	3.178e-03	4.917e-01	TNIP1:139 RPL28:224 PEPLO:234 MRLP2:372 SGRS1:450 RPDS5:541 SRBD1:2056
Regulation Of JNK Cascade (GO:0046328)	-0.08926013	93	3.018e-03	4.917e-01	MAP3K1:60 IGF1R:101 CYLD:133 WNT7B:470 TRAF6:497 HMGB1:680 SERPINF2:793
Regulation Of Type B Pancreatic Cell Pro	-0.38735568	5	2.962e-03	4.917e-01	SGKP2:677 PTPRN1:1221 NR4A1:1516 NR4A3:1537 NUPR1:2446 NA NA
Negative Regulation Of SMAD Protein Sign	0.37634528	5	3.556e-03	5.039e-01	SLC2A10:50 PBLD:158 VEPH1A:894 LRPI:1341 CLIP:5418 NA NA
Acid Secretion (GO:0046717)	0.31623173	7	3.766e-03	5.057e-01	SLC22A16:24 DRD3:837 SLC51B:931 SLC51A:939 GHRL:1547 SLC2A6A:4198 SLC9A3R1:7916
Positive Regulation Of Isotype Switching	0.23272398	13	3.676e-03	5.057e-01	HMCE5:63 TGFBI:269 PAX1P1:902 KMT5C:924 KMT5B:1369 SHLD2:1655 TP53BP1:2234

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
BLALOCK_ALZHEIMERS_DISEASE_DN	-0.06204481	766	8.308e-09	5.380e-05	PTGES3:10 CDK17:15 UBL3:233 PLK2:54 CSNK1G3:60 ATP5F1:81 PPP2R2B:85
REACTOME_REGULATION_OF_EXPRESSION_OF_SLI	-0.16876670	80	1.868e-07	6.050e-04	PSMA7:138 RNP51:142 USP33:180 PSDM7:184 RPL28:224 RPLP0:234 PABPC1:374
REACTOME_THE_ROLE_OF_GTSE1_IN_G2_M_PROGR	-0.21293218	46	5.935e-07	1.281e-03	PSMA7:138 PSDM7:184 TP53:297 UBB:399 SEM1:702 PSMF1:809 PSMC3:818
HOUNKPE_HOUSEKEEPING_GENES	-0.05424055	712	1.129e-06	1.828e-03	SPAG9:18 TBL1XR1:40 COX15:42 BAP1:45 AP3M1:53 TME2M1:59 ZRANB2:87
REACTOME_SIGNALING_BY_WNT	-0.11009639	162	1.419e-06	1.838e-03	PLCB3:51 KMT2D:75 PSMA7:138 VPS35:144 PRICKLE1:163 PRKG1:167 PSDM7:184
REACTOME_SIGNALING_BY_ROBO_RECEPTORS	-0.12836661	113	2.545e-06	2.354e-03	PSMA7:138 RNP51:142 USP33:180 PSDM7:184 PNLN1:202 RPL28:224 RPLP0:234
REACTOME_TRANSCRIPTIONAL_REGULATION_BY_R	-0.12112995	128	2.328e-06	2.354e-03	PHC1:73 KMT2D:75 SLC13:138 PSDM7:184 CEBX:223 UBB:399 SEM1:702 PSMF1:809
REACTOME_DISEASES_OF_SIGNAL_TRANSDUCTION	-0.07854398	297	6.613e-06	2.925e-03	RAP1A:1 POLR2E:21 TBL1XR1:40 MAP3K1:80 PSMA7:138 PSMF1:809 PSMC3:818
REACTOME_DEGRADATION_OF_GLI1_BY_THE_PROT	-0.23125302	32	6.023e-06	4.334e-03	PSMA7:138 PSDM7:184 UBB:399 SEM1:702 PSMF1:809 PSMC3:818
REACTOME_REGULATION_OF_MRNA_STABILITY_BY	-0.18948137	46	8.856e-06	5.735e-03	PSMA7:138 PSDM7:184 PABPC1:374 UBB:399 SEM1:702 PSMF1:809 PSMC3:818
MILI_PSEUDOPODIA_HAPTOTAXIS_UP	-0.07077197	328	1.182e-05	6.958e-03	RAP1A:1 AP3M1:53 CSNK1G3:60 PLEKHA8:84 ZRANB2:87 EIF2A:116 ARMC1:125
REACTOME_DEVELOPMENTAL_BIOLOGY	-0.04894056	681	1.700e-05	9.172e-03	NR5A2:12 SPAG9:18 POLR2E:21 EFNA5:29 TRPC5:31 GDNF:36 ITGA5:37
REACTOME_RUNX1_REGULATES_TRANSCRIPTION_O	-0.18262151	46	1.848e-05	9.206e-03	PSMA7:138 PSDM7:184 UBB:399 SEM1:702 PSMF1:809 PSMC3:818 PSMB10:1046
REACTOME_NEGATIVE_REGULATION_OF_NOTCH4_S	-0.23034815	27	3.449e-05	1.507e-02	PSMA7:138 PSDM7:184 UBB:399 SEM1:702 PSMF1:809 PSMC3:818
CAIRO_HEPATOBlastoma_DN	0.08602525	196	3.491e-05	1.507e-02	CPT2:59 TMEH131L:77 SLC1A1:84 ABCB1:144 ACA2:239 ACADM:307 DHTKD1:324
REACTOME_TCF_DEPENDENT_SIGNALING_IN_RESP	-0.11105458	115	4.022e-05	1.557e-02	KMT2D:75 PSMA7:138 PSDM7:184 KREMEN2:381 UBB:399 SEM1:702 PP2C8:806
REACTOME_ASMETRIC_LOCALIZATION_OF_PCP_	-0.19712965	36	4.292e-05	1.557e-02	PSMA7:138 PRICKLE1:163 PSDM7:184 UBB:399 PARDBA:642 SEM1:702 PSMF1:809
REACTOME_DEFECTIVE_CFTR_CAUSES_CYSTIC_FI	-0.20577752	33	4.326e-05	1.557e-02	PSMA7:138 PSDM7:184 UBB:399 SEM1:702 PSMC3:818 PSMB10:1046
REACTOME_MAPK_FAMILY_SIGNALING_CASCADES	-0.07968579	219	5.150e-05	1.755e-02	RAP1A:1 GDNF:36 MAP3K1:80 IL5RA:88 PSMA7:138 PSDM7:184 SEPTIN7:361
REACTOME_TRANSCRIPTIONAL_REGULATION_BY_R	-0.14034637	69	5.647e-05	1.829e-02	ITGA5:37 SMADE3:130 PSMA7:138 PSDM7:184 UBB:399 SEM1:702 PSMF1:809
REACTOME_REGULATION_OF_RAS_BY_GAPS	-0.18789741	38	6.173e-05	1.904e-02	PSMA7:138 PSDM7:184 UBB:399 SEM1:702 PSMF1:809 PSMC3:818 PSMB10:1046
REACTOME_AUF1_HNRNP_D0_BINDS_AND_DESTABI	-0.20721344	31	6.567e-05	1.933e-02	PSMA7:138 PSDM7:184 PABPC1:374 UBB:399 SEM1:702 PSMF1:809 PSMC3:818
REACTOME_SLG_MEDATED_TRANSMEMBRANE_TRAN	0.07801675	219	7.373e-05	2.076e-02	SLC5A12:15 SLC22A16:24 SLC2A10:50 SLC1A1:84 SLC1A2:162 SLC36A2:227 SLC2A11:264
HSIAO_HOUSEKEEPING_GENES	-0.08033602	204	8.084e-05	2.181e-02	ABLIM1:62 CD34:64 AARS1:170 PSDM7:184 PNLN1:202 RPL28:224 KARS1:226
REACTOME_SIGNALING_BY_THE_B_CELL_RECEPTO	-0.13658717	69	8.884e-05	2.301e-02	PSMA7:138 PSDM7:184 UBB:399 NFATC2:457 SEM1:702 PSMF1:809 PSMC3:818
REACTOME_ADAPTIVE_IMMUNE_SYSTEM	-0.05273123	470	1.026e-04	2.447e-02	RAP1A:1 ASB15:13 CD34:64 CD4:70 FBXW7:137 PSMA7:138 PRKG1:167
WP_NRF2_PATHWAY	0.11711941	92	1.058e-04	2.447e-02	SLC5A12:15 SLC2A10:50 TXNRD1:237 SLC2A11:264 UBB:399 SLC2A7:316 MGST3:343
KEGG_LYSINE_DEGRADATION	0.18256884	38	9.922e-05	2.447e-02	NSD2:72 EHADH1:16 EHM1:283 NSD3:352 GCDH:418 SETD1B:859 PIPOX:877
REACTOME_TRANSPORT_OF_SMALL_MOLECULES	0.04803619	566	1.100e-04	2.456e-02	TRPM6:6 TRPM5:12 SLC1A2:152 SLC22A16:24 ADCY1:27 ATP9B:38 TP8B2:47
REACTOME_TRANSPORT_OF_BILE_SALTS_AND_ORG	0.12714593	77	1.168e-04	2.521e-02	SLC22A16:24 RSC1A1:142 SLC1A2:332 SLC12A2:389 SLC4A4S:401 SLC6A6:519 SLC6A12:596
REACTOME_STABILIZATION_OF_P53	-0.18888309	34	1.390e-04	2.905e-02	PSMA7:138 PSDM7:184 TP53:297 MDM2:317 UBB:399 SEM1:702 PSMF1:809
KIM_ALL_DISORDERS_CALB1_CORR_UP	-0.06009307	337	1.638e-04	3.315e-02	POLR2E:21 UBL3:233 BAP1:45 PLK2:54 RTN4:133 TAP61VD:152 ADGRB2:153
REACTOME_SIGNALING_BY_NOTCH4	-0.15317167	49	2.099e-04	4.120e-02	PSMA7:138 PSDM7:184 UBB:399 TACC3:605 MAML3:614 SEM1:702 PSMF1:809
REACTOME_NEDDYATION	-0.08882418	146	2.191e-04	4.174e-02	ASB15:13 FBXW7:137 PSMA7:138 PSDM7:184 SPBS1:191 COP56:346 DCAF16:393
REACTOME_DOWNSTREAM_SIGNALING_EVENTS_OF_	-0.15888063	45	2.285e-04	4.228e-02	PSMA7:138 PSDM7:184 UBB:399 NFATC2:457 SEM1:702 PSMF1:809 PSMC3:818
REACTOME_UV_SPECIFIC_PROCESSING_PROTEASE	-0.10031610	111	2.677e-04	4.686e-02	USP19:103 CYLD:133 PSMA7:138 USP33:180 PSDM7:184 WDR20:240 TP53:297
REACTOME_REGULATION_OF_RUNX2_EXPRESSION	-0.16886861	39	2.650e-04	4.686e-02	PSMA7:138 PSDM7:184 UBB:399 SEM1:702 PSMF1:809 PSMC3:818 PSMB10:1046
REACTOME_PCP_CE_PATHWAY	-0.14570295	52	2.814e-04	4.795e-02	PSMA7:138 PRICKLE1:163 PSDM7:184 PNLN1:202 CLTC:348 UBB:399 PARDBA:642
ACEVEDO_LIVER_TUMOR_V5_NORMAL_ADJACENT_T	-0.04447811	579	2.947e-04	4.893e-02	AF1L:90 PKP2:97 PPP1CA:104 TBL2:109 PRRCC2:117 BVIM:129
REACTOME_SCF_BETA_TRCP_MEDIATED_DEGRADAT	-0.18959056	30	3.272e-04	5.105e-02	PSMA7:138 PSDM7:184 UBB:399 SEM1:702 PSMF1:809 PSMC3:818 PSMB10:1046

DisGeNET Top pathways by non-permutation

Calcium oxidate kidney stones	0.025709278	33	7.574e-03	6.763e-01	SLC6A2:227 AGXT:508 GRHRP:1735 SLC6A2:2794 OPLAH:2795 HOGA1:2899 SLC26A1:4104
Central neuroblastoma	-0.02894324	1086	1.788e-03	6.763e-01	RAP1A:1 NPPF:7 EDNRB:22 PMG1:24 GDNF:36 ITGAS:37 EBI3:48
Cerebral arterial aneurysm	0.17444013	57	5.079e-03	6.763e-01	TGFB1:269 ENG:467 MPMF2:514 COL12A2:50 SCN11A:547 TMEM132B:891 TNF:1025
Childhood Ependymoma	-0.24402082	10	7.502e-03	6.763e-01	TNCF:63 TP52:297 EBRB4:453 PIK3CD:1596 PIK3CG:2551 PIK3CA:3086 TP53NP1:4199
Chronic kidney disease stage 5	0.04750796	275	7.102e-03	6.763e-01	ENTPD1:46 CP72S:48 AB7:9 CHDH:49 CB8:120 TGFBI:269 TRPV1:295
Chronic Liver Failure	0.27018128	10	3.095e-03	6.763e-01	ABCB11:144 AKR1D1:628 IGFBR1:647 ABCB4:716 THPO:1404 EIF2AK3:2445 ABCB1:3529
Cilioc Epithelium	0.07255638	150	2.247e-03	6.763e-01	DYNC2H1:19 K1AA07D5:136 MICAL3:191 TGFBI:269 TMD1:280 ARL13B:284 CEP72:453
Classical Hodgkin's Lymphoma	-0.08887657	116	9.772e-04	6.763e-01	IGF1R:101 IL2:133 STAT5A:284 TP53:297 MDM2:317 IL10:428 CAP5:575
Cold intolerance	0.33845953	6	4.093e-03	6.763e-01	TRPV1:295 LPN1:986 UCP1:1123 TRPA1:1170 PARAA:1307 IRF4:7311 NA
Common wart	-0.35596474	5	5.843e-03	6.763e-01	TP53:297 MDM2:317 HSPA4:1227 CPK2A:2116 DCKN1A:5264 NA NA
Cone dysfunction syndrome	0.20705253	14	7.324e-03	6.763e-01	PROM1:80 RGS9:183 RGS9BP:574 PITPNM3:633 GPDH:1066 ADRGVL:1384 PAF:2543
Congenital exophthalmos	0.31345163	7	4.083e-03	6.763e-01	AFP:14 LRP1:1341 CHRNA7:1589 SDS:1829 SLC32A1:3565 GDNF:36 PSC5:4080
Congenital hemivertebra	-0.16190383	25	5.099e-03	6.763e-01	RAP1A:1 KMT2D:75 TMCO1:248 JAG1:960 ATX:1149 TBX6:1474 PUF60:2473
CREST Syndrome	-0.20310734	21	1.279e-03	6.763e-01	CCR1:241 IL10:428 TIMP1:464 HMGB1:660 RAB13:883 CCR6:854 JAK2:1277
Cystic Fibrosis	0.05044350	374	9.108e-04	6.763e-01	BPIFB1:1 MUC20:4 AF:14 CTRC:22 CELA1A:101 ENTPD1:46 ALB:79
Cystic Kidney Diseases	0.10695445	61	3.915e-03	6.763e-01	TCTN1:280 ARL13B:284 TCTN3:528 TSC1:929 CEP164:1018 NHPB:1082 TMEM67:1094
Decreased muscle mass	0.15792537	28	3.844e-03	6.763e-01	NSD2:72 HERC2:145 ATP6V0A2:150 SPG11:286 LETM1:383 HSD17B4:385 AP4E1:687

KeggReactome Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
TOME_TRANSPORT_OF_SMALL_MOLECULES	0.006089826	623	3.811e-07	6.871e-04	SLCSA12:4 TRPM6:5 CLCNKA:14 SLC11A2:17 CAMK2B:21 ADCY1:38 AZM:39
TOME_PEPTIDE_LIGAND_BINDING_RECEPTOR	-0.10230307	166	6.095e-06	5.949e-03	TACR3:39 GRP:50 NPPF:65 F2:170 APP:184 NMUR1:217 EDNRB:256
EACTOME_GLTUTHIONE_CONJUGATION	0.23843873	25	3.726e-05	1.933e-02	MGST3:74 GSTO1:338 CHAC2:355 F2:170 AKG1:869 MGST1:871 GSTZ1:1364 GGTS:1504
TOME_ION_CHANNEL_TRANSPORT	0.09253884	166	4.288e-05	1.933e-02	TRPM6:5 CLCNKA:14 CAMK2B:21 TRPM5:50 TRPV1:95 TRPM1:96 CLCNKB:140
TOME_ERYTHROCYTES_TAKE_UP_OXYGEN_AND	0.44967996	6	1.367e-04	4.107e-04	SLC4A1:156 RHAG:257 CA:408 CA2:608 AQP1:1236 KLC1:1740 NA
TOME_PROCESSING_OF_CAPPED_INTRON	0.25565218	19	1.153e-04	4.107e-02	SNRPD:239 SLBP:383 NUDT1:702 SYMPK:741 CP5F7:1206 SNRPG:1830 NCBP1:2395
TOME_MUSCLE_CONTRACTION	0.08240909	176	1.772e-04	4.564e-02	CAMK2B:21 KCNK15:11 ATP2B4:238 KCN1K6:296 ACTC1:332 MYL4:450 TPR1:501
TOME_SPERM_MOTILITY_AND_TAKES	0.36931054	8	2.994e-04	6.726e-02	CATSPERB:277 CATSPERG:820 CATSPERA:882 CATSPER2:1397 CATSPER3:1498 KCNU11:1620 CATSPER4:1740
TOME_STIMULI_SENSING_CHANNELS	0.10268056	101	3.789e-04	7.590e-02	TRPM6:5 CLCNKA:14 TRPM5:50 TRPV1:95 TRPM1:96 CLCNKB:140 TRPM7:172
TOME_RECYCLING_OF_BILE_ACIDS_AND_SAL	0.23589891	16	1.093e-03	7.612e-02	ALB:47 SLC01A2:237 SLC10A2:260 ABCB11:262 SLC01B1:558 SLC51A:924 NR1H4:1483
TOME_GENERATION_OF_SECOND_MESSENGER	-0.19365382	25	8.102e-04	7.612e-02	EVL:41 ZAP70:83 CD102:780 K2:802 CD247:807 HLA-DRA:1812 LCP2:957
TOME_GENE_SILENCING_BY_RNA	0.11167511	82	4.889e-04	7.612e-02	MOV10L1:46 H3-3A:194 POLR2J2:289 PWWL2:322 LRN:486 PWWL1:1536 NUP133:633
TOME_CLASS_A_1_RHODOPSIN_LIKE_RECEPT	-0.00029674	294	4.287e-04	7.612e-02	TACR3:39 GRP:50 NPPF:65 F2:170 APP:184 NMUR1:217 EDNRB:256
TOME_CLASS_C_3_METABOTROPIC_GLTUTHAM	0.03828153	27	9.881e-04	7.612e-02	TAS2R10:179 TAS2R81:862 GRM6:1156 TAS2R60:2042 TAS1R2:1208 TAS2R42:2250 TAS2R5:2250
TOME_TRANSPORT_OF_INORGANIC_CATIONS	0.09425009	101	1.105e-03	7.612e-02	SLCSA12:4 SLC6A6:142 SLC1A1:155 SLC1A1:320 SLC6A1:325 SLC28A3:412 SLC7A9:519
TOME_TRANSPORT_OF_NUTRIENTS	0.05522005	202	6.249e-04	7.612e-02	SLC6A1:325 SLC6A1:320 SLC6A1:320 SLC6A1:320 SLC6A1:320 SLC6A1:320 SLC6A1:320